

A constructive polyconvex neural constitutive surrogate for the nonlinear hyperelastic RVE

Satisfying – and correcting – the standard PANN admissibility checklist

July 29, 2026

Abstract

Physics-augmented neural networks (PANNs) for hyperelastic constitutive modelling are commonly held to a six-point admissibility checklist: (1) thermodynamic consistency, (2) objectivity, (3) tensor symmetry, (4) normalization at the reference configuration, (5) polyconvexity, and (6) a growth/coercivity condition [4, 1]. This memorandum works through each point in full mathematical detail for a single certified model of the two-dimensional nonlinear hyperelastic representative volume element (RVE) treated in this project, an energy-based PANN built from an input-convex neural network (ICNN) acting on balanced directional invariants of F and $\text{cof } F$. Five of the six points are satisfied exactly, by construction, with a closed-form proof given below for each. The sixth – polyconvexity – is where a common informal shorthand in the literature and in internal seminar material, “ $\partial\sigma/\partial\epsilon$ is symmetric positive definite (SPD)”, is *not* what polyconvexity means, is not implied by it, is not implied by objectivity-respecting hyperelasticity in general, and – as we show directly from this project’s own full-order model (FOM) tangent audit – is not even true of the ground truth being learned. We give Ball’s actual definition, prove our model satisfies it by an explicit construction (not a numerical penalty), explain precisely which weaker condition (rank-one/Legendre–Hadamard convexity) polyconvexity does imply and that we do verify numerically, and show why demanding the stronger SPD condition would force the surrogate to *contradict* the FOM. The model is trained on Stage–1 trajectories only, evaluated once on the untouched Stage–10 trajectory, and compared against a projection-based hyper-reduced-order model (HPROM–ANN, in both its iterative-corrector and direct-evaluation operating modes) on the same path, and against a direct strain-to-stress regression baseline with no energy potential at all, which is shown to violate thermodynamic consistency directly (a nonzero cyclic-work integral around closed strain loops) despite matching or exceeding every certified model’s held-out accuracy. A state-of-the-art survey of seven directly relevant physics-augmented neural network (PANN) constructions is given, precisely comparing which admissibility properties each achieves exactly, approximately, or not at all; it identifies, in the surveyed authors’ own words, an explicit open problem – combining exact objectivity, an exact stress-free reference state, and exact polyconvexity for an anisotropic material with no assumed symmetry group – that the construction here resolves for this RVE. The same proofs are shown to transfer verbatim to a second convex core, an input-convex Kolmogorov–Arnold network, at comparable accuracy with far fewer parameters; and a matched, otherwise identical, uncertified flexible PANN is used to show directly that held-out accuracy cannot detect the absence of a polyconvexity certificate – it matches or beats the certified models on every accuracy metric while failing a rank-one convexity check on 7% of sampled, ordinary deformation states.

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1 The admissibility checklist

A recurring checklist in the PANN literature and in this project’s own working seminar material requires an admissible hyperelastic surrogate W , with $\boldsymbol{\sigma} = \partial W / \partial \boldsymbol{\varepsilon}$, to satisfy:

- (C1) **Thermodynamic consistency** (reduced Clausius–Duhem inequality): $\boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$.
- (C2) **Objectivity**: $W(\boldsymbol{\varepsilon}) = W(\mathbf{R}\boldsymbol{\varepsilon})$ for every admissible rigid rotation of the current configuration.
- (C3) **Symmetry**: $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}) = \boldsymbol{\sigma}(\boldsymbol{\varepsilon})^T$.
- (C4) **Normalization**: $\boldsymbol{\sigma}(\mathbf{0}) = \mathbf{0}$ and $W(\mathbf{0}) = 0$.
- (C5) **Polyconvexity**: often shorthanded in seminar slides as “ $\mathbf{D} = \partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon}$ is SPD”.
- (C6) **Growth condition**: $W \rightarrow \infty$ as $\boldsymbol{\varepsilon} \rightarrow \infty$ (with extra terms if needed).

Sections 4–10 take each point in turn. (C1)–(C4) and (C6) are satisfied exactly, by construction, independent of training. (C5) requires the most care: Section 9 shows that the informal shorthand above is not Ball’s polyconvexity, is strictly stronger than it, and is in fact false for this RVE’s own ground-truth response – and then gives the correct definition, the correct (weaker) necessary condition we actually falsify-test, and the explicit construction that proves our model polyconvex in Ball’s sense. Before doing so, Section 2 places this checklist against the current physics-augmented neural network (PANN) literature: which parts are already correctly and rigorously solved elsewhere, which published claims are imprecise or documentedly incorrect, and which combination of requirements – specifically, a *general* anisotropic material with no assumed symmetry group – remains an open problem that this work targets.

2 Related work: what is proven, what is imprecise, and what remains open

This section surveys the seven works of Table 1, plus several more specific supporting references, chosen because each makes an explicit, checkable claim about one or more of (C1)–(C6) for a neural-network hyperelastic energy. Table 1 summarizes the seven-work comparison; the discussion below justifies every entry

and gives the precise textual basis for each claim, since “X satisfies polyconvexity” is stated in nearly all of this literature and the differences are entirely in *which* polyconvexity-adjacent condition is actually being claimed.

Table 1: Comparison against the admissibility checklist. “Exact” means satisfied by construction for every parameter value, not approximated by training or data augmentation. “Group-free anisotropy” means the model represents a general anisotropic material without committing to a named symmetry class (isotropic, transversely isotropic, cubic, orthotropic, ...) or a learned member of one. “Growth exercised” means the model’s volumetric term actually diverges/was actually tested at $J \rightarrow 0^+$, as opposed to being stated as a requirement but bypassed because the worked examples are (nearly) incompressible.

Work	Exact obj.	Exact ref. state	Ball polyconvex	Group-free anisotropy	Growth exercised
As’ad et al. [5]	yes	yes	no (E-convex)	yes (raw $\mathbf{E}$)	n/a
Linka & Kuhl [6]	yes	yes	yes	no (isotropic)	no (incompressible)
Tac et al. [7]	yes	partial ¹	yes	no (2 fixed fibers)	no (incompressible)
Klein et al. [4]	yes/approx. ²	approx. only	yes	no (named group)	yes
Thakolkaran et al. [8]	yes	yes	yes	no (isotropic)	no (no barrier term)
Linden et al. [9]	yes	yes	yes	no (named group)	yes
Kalina et al. [10]	yes	yes	no	partial ³	yes
This work	yes	yes	yes	yes	yes

2.1 Models that enforce a strictly stronger condition than polyconvexity

As’ad, Avery and Farhat [5] build a hyperelastic PANN for an anisotropic woven-fabric RVE homogenized with the same kind of FE² pipeline used in this project, and validate it inside an actual coupled fluid–structure simulation (the supersonic inflation of a parachute canopy) – the only work surveyed here that demonstrates a downstream, large-scale numerical-robustness benefit, not just held-out accuracy. Their own Section 3.3 states the reason plainly: “enforcing polyconvexity in F is difficult as E is not in general convex in F . On the other hand, consider the convexity of W in E , which is equivalent to... $\mathbb{C} = \partial \mathbf{S} / \partial \mathbf{E} \succeq 0$.” This is, verbatim, the informal shorthand of (C5) in Section 1: a global convex ICNN is applied directly to the raw Green–Lagrange strain, which by elementary calculus forces $\partial \mathbf{S} / \partial \mathbf{E} \succeq 0$ *everywhere*, not merely at the reference state. The authors are explicit that this is a deliberate substitute for polyconvexity, not an oversight, and they give a real justification for it: their own Eq. (13) decomposes the F -Hessian of W into a material-stiffness term (made non-negative by convexity in E) and a geometric-stiffness term built from the ambient stress $\mathbf{S}$ itself, which can be negative in compression – so ellipticity in F (and hence buckling-type non-uniqueness) can still be lost even though $\partial \mathbf{S} / \partial \mathbf{E} \succeq 0$ holds globally. This is a legitimate and carefully reasoned engineering trade-off, and their finite-element examples (Section 4.4) show it buys real numerical robustness: Newton’s method diverges for a non-convex ANN under load, and converges for their convex-in- $\mathbf{E}$ one.

It is nonetheless exactly the condition Section 9.1 argues against for a material like the one in this project. Unlike convexity in F itself – which As’ad et al.’s own Section 3.3 correctly notes is incompatible with objectivity, citing Coleman and Noll’s classical counterexample – convexity in $\mathbf{E}$ raises no such contradiction, since $\mathbf{E}$ (equivalently C) is already an objective strain measure; Gao, Neff, Roventa and Thiel [11] build a valid existence theory on exactly this (stronger-than-polyconvex) hypothesis, so it is a coherent, usable condition in general, not an impossible one. The reason to avoid it for this RVE is not a general impossibility but a specific, measured fact: the FOM tangent audit gives a negative eigenvalue, -0.0648 GPa, at a Stage-10 state, so $\partial \mathbf{S} / \partial \mathbf{E} \succeq 0$ is simply false for this material’s true response, and a convex-in- $\mathbf{E}$ ICNN applied to this RVE would be structurally unable to reproduce that state. Two further, more mechanical differences follow directly from this choice. First, As’ad et al.’s stress-free-reference correction is a fully general linear term $\mathbf{h} \cdot \mathbf{E}$ with $\mathbf{h} = -\partial \theta / \partial \mathbf{E}(\mathbf{0})$ (their Eqs. 15–16): adding *any* linear function of $\mathbf{E}$ preserves convexity in $\mathbf{E}$, so no special structure among the three strain components is required. A linear-in- $\mathbf{E}$ correction is a *quadratic*, generally non-convex-in- F form once expressed in F , so this shortcut is not available to a polyconvex-in- F construction; this is precisely why Proposition 2 needs the balanced-direction identity (16) instead of a free 3-parameter correction. Second, As’ad et al.’s ICNN needs only the classical Amos et al. convexity-only construction (unconstrained input-to-hidden weights), because it is applied directly to the raw strain; our construction needs the strictly stronger *monotone* ICNN (non-negative weights at every layer, Section 9.3)

¹Exact for their CANN/NODE cores; an unaddressed gap for their ICNN core, whose biases are unconstrained.

²Exact for their invariant model W^I ; only approximated by data augmentation for their deformation-gradient model W^F .

³Anisotropy *parameters* (e.g. fiber directions) within a pre-chosen symmetry class are learned; the class itself is still user-specified.

because the ICNN here is composed with an intermediate layer of engineered convex features of F , and lifting convexity through that composition requires the outer function to be coordinate-wise non-decreasing. Finally, their default loss normalizes every stress component by its own per-component standard deviation (their Eq. 2) – i.e. always at the extreme $0.00 = 1$ of (20) – which Table 5 shows costs a factor of roughly two in von Mises accuracy for an RVE whose components differ in magnitude by two orders of magnitude; this trade-off is not discussed in [5], most likely because their fabric material’s stress components do not differ as severely.

The same global-SPD-tangent condition, by a different mechanism, underlies Xu, Huang and Darve [12]: rather than an ICNN, their SPD-NN predicts the Cholesky factor of the tangent stiffness directly, so that the stiffness matrix is symmetric positive definite by construction and stress is integrated from it incrementally. This is applied to anisotropic fiber-reinforced plates and to path-dependent (elasto-plastic) response, which the ICNN constructions surveyed here do not attempt, at the cost of the same strictly-stronger-than-polyconvex commitment as Section 2.1, for the same reason: a positive-definite tangent stiffness *is* $\mathbf{D} \succeq 0$, restated in rate form.

2.2 Models that enforce Ball’s polyconvexity, but only for a named symmetry group

Klein et al. [4] (already used throughout Section 9 as the primary reference for the correct definition) give two ICNN constructions, both restricted to a material symmetry group $\mathcal{G} \subset SO(3)$ specified a priori (cubic, transversely isotropic) via a structural tensor. Their invariant-based model W^I gets objectivity, material symmetry, and polyconvexity exactly, but not the reference state: “the model (16) is not stress-free by construction ... this may thus lead to a loss of ellipticity.” Their own explanation for why is worth quoting in full, because it is the state of the art this project improves on: “In Fernández et al. (2021b), a projection approach is proposed which fulfills the stress-free reference configuration in an exact way. While this approach preserves polyconvexity when formulated in terms of F instead of C , it is not compatible with the methods of incorporating objectivity and material symmetry which are used in this work, and therefore cannot be applied.” Their deformation-gradient model W^F makes the opposite trade: exact polyconvexity and material symmetry, but objectivity is only *approximated* by data augmentation over 1024 random rotations (their Eq. 22), which they candidly report requires 64 observers before the shear response becomes acceptably tight (their Table 1, Fig. 4).

Linden et al. [9] close this specific gap for isotropic and transversely isotropic materials: they construct an additive stress- and energy-normalization term, restricted to J and the symmetry-respecting anisotropic invariants (never the raw off-diagonal components of C), and show this achieves exact objectivity, exact reference state, exact polyconvexity, and exact material symmetry *simultaneously* – solving the open problem Klein et al. identified. Critically, however, they do this only within a symmetry class fixed by a structural tensor, and they are explicit that this does not extend to a general material: “there are symmetry groups for which no complete set can be found that does not violate polyconvexity ... it might be possible that there does not exist a set of invariants to describe the anisotropy of materials with arbitrary microstructures, e.g., 3D-printed composites.” This is, in the words of the researchers who solved the harder half of this problem, an open question for exactly the case this memorandum addresses: an RVE with no known or assumed symmetry class. The mechanism used here answers it for that case: the balanced, non- D_4 -closed direction systems of Table 4 are not tied to any structural tensor or symmetry group, and Proposition 1 shows they still produce the isotropic reference-derivative structure that an exact, polyconvexity-preserving correction requires.

2.3 A model that widens the anisotropy class by giving up polyconvexity

Kalina et al. [10] take the opposite trade from Section 2.2: they still require a user-specified symmetry class (one of 14 crystallographic groups), but the structure tensor’s own parameters (e.g. fiber directions) within that class are learned jointly with the network from data, which is a substantially more flexible anisotropy treatment than a fixed structural tensor. In exchange, they explicitly decline polyconvexity: “polyconvexity can be too restrictive, especially for multiscale modeling,” relying instead on physics-augmented correction terms in the loss rather than an architecturally convex core. Together with Section 2.2, this is the clearest evidence in the current literature for a real trade-off frontier: the more general the anisotropy representation gets, the harder (and, so far, unachieved outside this work) it has been to keep an exact, architectural polyconvexity certificate.

2.4 Isotropic and fiber-restricted polyconvex CANNs, and the growth condition in practice

Linka & Kuhl [6], Tac et al. [7], and Thakolkaran et al. [8] all correctly use the additive, block-decoupled sufficient condition for polyconvexity (convex, non-decreasing scalar functions of I_1 , of a polyconvex modification of I_2 , and of J , summed), and Tac et al. are unusually careful to state that this is strictly stronger than what is needed and to distinguish it explicitly from rank-one convexity: “[r]ank-one convexity [is] weaker than polyconvexity and not sufficient for the existence of global minimizers.” None of the three, however, is anisotropic beyond at most two fixed fiber directions (transverse isotropy), and the growth condition (C6) is uniformly *stated but not exercised*: Linka & Kuhl and Tac et al. both work with perfectly or nearly incompressible materials, so $J = 1$ is enforced by a Lagrange-multiplier pressure and the stated $J \rightarrow 0^+$ divergence is never actually active in a trained model. Thakolkaran et al.’s ICKAN paper is, on the polyconvexity definition itself, the most careful of the three – it states Ball’s condition precisely and does not conflate it with anything stronger – but it uses $K_3 = (J - 1)^2$ as its sole volumetric term, which is finite (equal to 1) at $J = 0$: there is no divergent barrier at all, and (C6) is not satisfied by their model as published. Our own volumetric term is exercised, not merely stated: the audit in Table 3–4 and Figure 1 samples W_{pc} down to $J = 10^{-6}$ and up to $J = 100$ for the trained model, for a compressible, general anisotropic material, which is a more demanding setting than any of the three incompressible or bounded-volumetric examples above.

Two further isotropic-only works round out this picture, at opposite ends of the convexity-vs-flexibility trade-off already seen in Section 2.3. Tac, Sahli-Costabal and Buganza Tepole [13] originate the neural-ODE route to the same additive, monotone-in-the-invariants sufficient condition used in [6, 7], demonstrated on anisotropic (two-fixed-fiber) skin data but, like the others in this subsection, without an exercised $J \rightarrow 0^+$ barrier. Abdolazizi, Aydin, Cyron and Linka [14], by contrast, apply a plain, unconstrained Kolmogorov–Arnold network to the same isotropic invariants and explicitly decline any convexity requirement at all: objectivity, material symmetry, and the reference state are enforced by construction, but polyconvexity, rank-one convexity, and growth are not discussed as design goals anywhere in the paper. Their examples (rubber, brain tissue, silicone) show good fits with no reported instability, which is consistent with, not evidence against, the point made throughout this section: losing the polyconvexity certificate does not necessarily show up as a visible failure on in-distribution calibration data, which is precisely why it needs to be guaranteed by construction rather than checked post hoc.

2.5 Where published claims have been documented to be wrong

Beyond the informal seminar shorthand this memorandum sets out to correct, the literature itself documents concrete, checkable errors of exactly this kind, which is useful context for how easy this is to get wrong even with a stated intent to be rigorous. Klein et al. [4] report that Ghaderi et al. [15] claim a polyconvex ANN potential but use hyperbolic-tangent activations, which are not convex, so “the potential proposed in Ghaderi et al. (2020) violates the polyconvexity condition.” We checked this directly against the original rather than taking it on trust: Ghaderi et al. do state, in their own Appendix B.1, that “[i]f weights, which connect the input of λ_j to other neurons, are positive, the proposed model holds the condition of polyconvexity,” and their Section 4 specifies the network core uses “soft plus . . . , sinusoid . . . and hyperbolic tangent” activations together. Weight non-negativity alone does not make a composition convex if the activation itself is not convex and non-decreasing (Section 9.3); sine is neither, and tanh is concave, not convex, on the positive axis. Klein et al.’s report is accurate, and the error is exactly the kind Section 9.3 is built to avoid by construction: stating a sufficient condition on the weights while leaving out that it also requires convex, non-decreasing activations. They report the same failure mode for Tac et al. (2021), whose model uses the invariant $\bar{I}_2 = \frac{1}{2}(\text{tr } \bar{C})^2 - \frac{1}{2}\text{tr } \bar{C}^2$ as a direct network input: this particular invariant is *not* polyconvex (Hartmann and Neff, cited by Klein et al. as their Lemma 2.4), so despite the stated intent, “the polyconvexity condition is violated by the choice of input arguments.” Both are examples of the same lesson emphasized in Section 9.3: polyconvexity is a property of the specific functional form all the way down to the choice of activation function and input invariant, not a label that can be attached to an architecture in general, and a single non-convex activation or a single non-polyconvex invariant among otherwise-correct ones is enough to void the entire certificate.

2.6 What remains open, and what this work contributes

Reading Sections 2.1–2.5 together, the state of the art as of this writing has, to our knowledge, not produced a model that is simultaneously (i) exactly objective, (ii) exactly stress-free and energy-free at the reference state, (iii) exactly polyconvex in Ball’s sense, and (iv) anisotropic without committing to a named symmetry group – the combination Linden et al. [9] explicitly flag as possibly requiring an invariant set that does

not exist for a general microstructure. This memorandum’s contribution is a specific, checkable answer for the case of a 2D RVE: replace the symmetry-group structural tensor with several *generic*, non- D_4 -closed direction systems, each individually balanced (Table 4), so that Proposition 1’s isotropic-reference-derivative argument goes through without ever assuming or checking which symmetry class, if any, the material belongs to. A secondary, smaller contribution is the loss-design result of Section 11: none of the seven works surveyed above discusses the possibility that per-component loss normalization (used, in its most extreme form, by [5], and in a milder blended form by 0.65 of the previously certified version of this model) can measurably *hurt* accuracy on the physically relevant equivalent stress when the material’s stress components differ by orders of magnitude, purely because it forces capacity to be spent chasing the smallest, noisiest component’s relative error. Finally, a limitation this work shares with every polyconvex-PANN paper surveyed except As’ad et al.: none of Sections 4–13 demonstrate that the polyconvexity certificate translates into the kind of downstream finite-element robustness (guaranteed Newton convergence, bounded transient kinetic energy) that [5] demonstrates for their strictly stronger, E-convex construction. Embedding this certified polyconvex PANN in an actual structural or multiscale solve, and checking whether the same robustness benefits appear without requiring the stronger and, for this material, false SPD-tangent assumption, is the natural next step to make this defensible end-to-end rather than only at the level of the standalone constitutive surrogate.

3 Kinematics and stress convention

Let F be the deformation gradient, and

$$C = F^T F, \quad E = \frac{1}{2}(C - I), \quad J = \det F = \sqrt{\det C} > 0. \quad (1)$$

The database and the model use the engineering-shear coordinate

$$\boldsymbol{\varepsilon} = \begin{bmatrix} E_{11} \\ E_{22} \\ \gamma_{12} \end{bmatrix}, \quad \gamma_{12} = 2E_{12}, \quad (2)$$

so that the work identity

$$dW = S_{11} dE_{11} + S_{22} dE_{22} + S_{12} d\gamma_{12}, \quad \boldsymbol{\sigma} := \frac{\partial W}{\partial \boldsymbol{\varepsilon}} = \begin{bmatrix} S_{11} \\ S_{22} \\ S_{12} \end{bmatrix} \quad (3)$$

returns exactly the second Piola–Kirchhoff components stored in the direct FOM data, with no factor-two ambiguity in the shear term and no independently trained stress network: every stress reported in this document is $\partial W / \partial \boldsymbol{\varepsilon}$ of the same scalar W .

4 (C1) Thermodynamic consistency

For a purely elastic RVE with no internal (dissipative) state variables, the reduced isothermal Clausius–Duhem inequality is $\mathcal{D} := \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$. Since $W = W(\boldsymbol{\varepsilon}(t))$ and W is differentiable,

$$\dot{W} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}} \cdot \dot{\boldsymbol{\varepsilon}} = \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} \quad (4)$$

identically, for *any* smooth scalar potential W and *any* strain history $\boldsymbol{\varepsilon}(t)$, by the chain rule alone. Hence $\mathcal{D} \equiv 0 \geq 0$: the inequality holds with equality, and the model is thermodynamically consistent (fully reversible, zero dissipation) for the same reason every hyperelastic (Green-elastic) model is – because the stress is defined as $\partial W / \partial \boldsymbol{\varepsilon}$ in (3) and not as an independently parameterized function. This is not a property that training can violate or that needs to be checked numerically; it is a property of the functional form (3) itself.

5 A regression-ANN baseline: violating (C1) directly

Section 4 shows (C1) holds for *any* model of the form $\boldsymbol{S} = \partial W / \partial \boldsymbol{\varepsilon}$, regardless of which W is learned. It says nothing about a model that is not of this form. As’ad et al. [5] motivate their own hyperelastic-ANN construction by first exhibiting exactly such a model – a plain regression network that maps strain directly to stress, with no potential anywhere – as their baseline “regression ANN”, the bottom tier of a three-tier ablation (regression ANN, hyperelastic ANN, convex hyperelastic ANN) that motivates each of their architectural commitments in turn. This section repeats that ablation for this project’s own RVE and its own admissibility checklist, adding a fourth tier below the two certified polyconvex cores and the uncertified free baseline of Section 15:

- **Tier 1 (this section):** a direct regression MLP, widths (128, 128, 64), mapping normalized $[E_{11}, E_{22}, \gamma_{12}]$ straight to normalized $[S_{11}, S_{22}, S_{12}]$ – **stress()** is a forward pass, never a gradient. No energy functional exists in this model at all.
- **Tier 2:** the free hyperelastic PANN of Section 15 – an unconstrained energy W_θ , so (C1) holds, but nothing else in (C4)–(C6) is guaranteed.
- **Tiers 3a/3b:** the certified polyconvex ICNN and ICKAN of Sections 9.3 and 14, satisfying all six checklist items exactly.

A direct, quantitative test of (C1). For a genuine hyperelastic model, integrating $\dot{W} = \mathbf{S} \cdot \dot{\boldsymbol{\varepsilon}}$ (Section 4) along any closed strain loop $\mathcal{L}$ – a path in strain space with the same start and end point – gives

$$\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon} = \oint_{\mathcal{L}} dW = W(\boldsymbol{\varepsilon}_{\text{end}}) - W(\boldsymbol{\varepsilon}_{\text{start}}) = 0, \quad (5)$$

exactly, for *every* closed loop, independently of which admissible W produced $\mathbf{S}$: this is a direct corollary of (C1), not a new physical assumption. For the tier-1 regression model, $\mathbf{S}$ is not the gradient of anything, so nothing in its construction forces (5), and there is no reason to expect it numerically. We evaluate (5) by trapezoidal quadrature (200 points per edge) on four closed polygonal loops entirely inside the Stage-1 training range (two axis-aligned normal-strain squares of different size, a shear loop, and an irregular triangle mixing normal and shear strain), for all four tiers, using each model’s own stress output (a direct forward pass for tier 1, $\partial W_\theta / \partial \boldsymbol{\varepsilon}$ by automatic differentiation for tiers 2–3b).

Table 2: Cyclic work $\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon}$ (Pa) around four closed strain loops, trapezoidal quadrature, 200 points per edge. Exactly zero is expected, by (5), for tiers 2–3b; nothing constrains tier 1.

Loop	Regression (tier 1)	Free (tier 2)	ICNN (tier 3a)	ICKAN (tier 3b)
Small normal square	$-3.042e + 04$ Pa	$3.024e + 00$ Pa	$-1.490e - 08$ Pa	$0.000e + 00$ Pa
Large normal square	$-6.687e + 04$ Pa	$4.128e + 00$ Pa	$0.000e + 00$ Pa	$1.192e - 07$ Pa
Shear loop	$-3.583e + 04$ Pa	$8.769e + 00$ Pa	$-2.757e - 07$ Pa	$7.451e - 09$ Pa
Mixed triangle	$9.850e + 03$ Pa	$-4.771e + 01$ Pa	$-4.968e + 01$ Pa	$-4.922e + 01$ Pa

For context, a typical energy density for this RVE (the training-set energy scale used to normalize the loss) is $2.136e + 09$ Pa: the tier-1 residual on the two normal-strain loops and the shear loop is five to six orders of magnitude smaller than this scale (0.003% of it at worst) – small in absolute terms, but, as the next paragraph shows, not a quadrature artifact, unlike the superficially similar-looking residuals of tiers 2–3b on the mixed triangle loop.

Distinguishing a real violation from quadrature error. The mixed-triangle residuals of tiers 2–3b in Table 2 are not exactly zero at 200 points per edge, because the trapezoidal rule only approximates a line integral of a nonlinear gradient field; but (5) holds exactly in the continuum limit, so this residual must vanish as the quadrature is refined, regardless of which admissible W produced $\mathbf{S}$. The tier-1 residual has no such reason to vanish, because it is not quadrature error in the first place – it is the trapezoidal estimate of a line integral of a non-conservative vector field, which is generally path-dependent and nonzero even in the continuum limit. Refining the same *mixedtriangle* loop by $4\times$ and $16\times$ (200, 800, 3200 points per edge) confirms exactly this:

- Regression (tier 1): $9.850e + 03$ Pa, $9.900e + 03$ Pa, $9.902e + 03$ Pa – stable, non-vanishing.
- Free (tier 2): $-4.771e + 01$ Pa, $-3.160e + 00$ Pa, $9.079e - 02$ Pa – shrinking toward zero.
- ICNN (tier 3a): $-4.968e + 01$ Pa, $-3.105e + 00$ Pa, $-1.940e - 01$ Pa – shrinking toward zero.
- ICKAN (tier 3b): $-4.922e + 01$ Pa, $-3.075e + 00$ Pa, $-1.921e - 01$ Pa – shrinking toward zero.

The three energy-based tiers converge to zero at the rate expected of trapezoidal quadrature error for a smooth integrand ($\mathcal{O}(h^2)$, i.e. roughly a $16\times$ reduction per $4\times$ refinement); the tier-1 residual instead converges to a fixed nonzero constant. This is not a difference of degree: it is the signature of a genuine curl in the tier-1 stress field, i.e. a direct, measured violation of (C1), as distinct from finite-sample noise.

Accuracy does not reveal this violation – if anything, it hides it. Table 6 reports the tier-1 model’s own held-out Stage-10 accuracy: 0.054% overall stress error and 0.033% von Mises error, *better than every other tier in this study*, including both certified polyconvex cores. This is the sharpest version yet of the point made in Section 15 with the free baseline: an aggregate accuracy metric on a held-out trajectory not only fails to detect the absence of a physical constraint, it can reward the model that discards the most physics, because a direct regression has no architectural constraint competing with the stress fit at all. The cyclic-work test of Table 2 is not an alternative way of measuring the same thing accuracy already measures; it is the only test in this memorandum that checks (C1) itself; direct observation, not a metric to be simultaneously optimized.

6 (C2) Objectivity

Objectivity concerns a superposed rigid rotation $Q \in \text{SO}(2)$ of the *current* configuration, $F \mapsto QF$. Since

$$(QF)^T(QF) = F^T Q^T Q F = F^T F = C, \quad (6)$$

any scalar built only from C (equivalently, from $F^T F$, $\text{cof}(F)^T \text{cof}(F) = \text{cof}(C)$, and $\det F$, as in Section 9) automatically satisfies $W(QF) = W(F)$ for every $Q \in \text{SO}(2)$. Every feature used by the model in Section 9 is built exclusively from C and $\text{cof}(C)$, so objectivity holds identically, not approximately, and not because it was included as a training penalty. This is different from *material* symmetry, which concerns rotating the RVE relative to its own fixed material axes and is discussed next.

7 (C3) Symmetry: two different meanings

The word “symmetry” is used for two unrelated properties in continuum mechanics, and the checklist item (C3) refers to the first, not the second:

- **Tensor symmetry**, $\sigma = \sigma^T$: a basic consequence of balance of angular momentum in the absence of body couples. It must hold for *any* admissible constitutive model, including anisotropic ones.
- **Material symmetry**: invariance of W under a *material* symmetry group (e.g. the D_4 group of a square lattice), $W(C_{11}, C_{22}, C_{12}) = W(C_{22}, C_{11}, -C_{12})$. This is a *modelling choice*, not a conservation law, and this project deliberately does *not* impose it: the RVE is treated as a general anisotropic material, because a square-shaped numerical cell does not by itself make the effective homogenized material isotropic or D_4 symmetric.

Tensor symmetry (the actual content of C3) holds identically in this model: the second Piola stress is parameterized by exactly the three independent numbers (S_{11}, S_{22}, S_{12}) of (3), which are then placed into the symmetric matrix $\begin{bmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{bmatrix}$. There is no separate, independently learned S_{21} that could disagree with S_{12} : $\sigma = \sigma^T$ by the structure of the parameterization, not as an approximate or penalized identity. Material symmetry (D_4) is a different, optional property that this model does not claim.

8 (C4) Normalization: the stress-free reference state

8.1 The energy value: trivial by construction

Let Φ_θ be a scalar network of a feature vector $\mathbf{z}_{\text{pc}}(F)$ defined in Section 9, and let $\mathbf{z}_{\text{pc}}(I)$ denote the features evaluated at the reference state $F = I$. The energy is defined as

$$W_{\text{pc}}(F) = \Phi_\theta(\mathbf{z}_{\text{pc}}(F)) - \Phi_\theta(\mathbf{z}_{\text{pc}}(I)) - r \log J + \frac{\beta}{2}(J - 1)^2. \quad (7)$$

At $F = I$: $\mathbf{z}_{\text{pc}}(F) = \mathbf{z}_{\text{pc}}(I)$ so the first two terms cancel exactly; $J = 1$ so $\log J = 0$ and $(J - 1)^2 = 0$. Hence $W_{\text{pc}}(I) = 0$ *exactly*, for *any* network Φ_θ and *any* parameters θ , before any training happens. This part of (C4) requires no special argument about the structural directions.

8.2 The stress value: not trivial, and this is where the balanced directions matter

Differentiating (7) and evaluating at $\varepsilon = 0$ is not automatically zero: it requires the gradient of the Φ_θ term to be exactly cancelled by the $-r \log J$ term, which only has an isotropic (equal-normal, zero-shear) shape. We now show this cancellation in full.

Lemma 1 (Isotropic reference derivative). *Let d be a unit vector in $\mathbb{R}^2$. Then $\text{cof}(d \otimes d) = I - d \otimes d$.*

Proof. For $d = (\cos \theta, \sin \theta)$, $d \otimes d = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$, and by the 2×2 cofactor formula $\text{cof}(d \otimes d) = \begin{bmatrix} \sin^2 \theta & -\cos \theta \sin \theta \\ -\cos \theta \sin \theta & \cos^2 \theta \end{bmatrix} = I - d \otimes d$, using $\cos^2 \theta + \sin^2 \theta = 1$. $\square$

Proposition 1 (Every feature has an isotropic C -gradient at $C = I$). *Let d_{k1}, d_{k2}, d_{k3} be unit vectors and $w_{k1}, w_{k2}, w_{k3} > 0$ satisfy the balance condition $\sum_i w_{ki} d_{ki} \otimes d_{ki} = I$ for each system $k = 1, 2, 3$, and set (Section 9)*

$$I_1 = \text{tr } C, \quad Q_{k,p}^F = \sum_i w_{ki} (d_{ki} \cdot C d_{ki})^p, \quad Q_{k,p}^H = \sum_i w_{ki} (d_{ki} \cdot \text{cof}(C) d_{ki})^p, \quad p = 2, 3.$$

Then at $C = I$, every one of these $1 + 12 = 13$ scalar features, together with J and J^2 , has a C -gradient proportional to the identity matrix, with the explicit coefficients in Table 3.

Proof. Taking the trace of the balance identity, $\sum_i w_{ki} |d_{ki}|^2 = \text{tr}(I) = 2$ and $|d_{ki}| = 1$ gives $\sum_i w_{ki} = 2$ for every system k (matching the numerical weights recorded in Table 4).

I_1 : $dI_1/dC = I$ exactly, for every C .

$Q_{k,p}^F$: $dQ_{k,p}^F/dC = \sum_i p w_{ki} (d_{ki} \cdot C d_{ki})^{p-1} d_{ki} \otimes d_{ki}$. At $C = I$, $d_{ki} \cdot I d_{ki} = |d_{ki}|^2 = 1$, so $dQ_{k,p}^F/dC|_{C=I} = p \sum_i w_{ki} d_{ki} \otimes d_{ki} = pI$.

$Q_{k,p}^H$: since $d \cdot \text{cof}(C) d = \text{cof}(C) : (d \otimes d)$ and the cofactor operation is self-adjoint under the Frobenius product for 2×2 matrices ($A : \text{cof}(B) = B : \text{cof}(A)$, verified by direct expansion), $d \cdot \text{cof}(C) d = C : \text{cof}(d \otimes d)$, which is linear in C with constant gradient $d(d \cdot \text{cof}(C) d)/dC = \text{cof}(d \otimes d) = I - d \otimes d$ by Lemma 1. Hence $dQ_{k,p}^H/dC = \sum_i p w_{ki} (d_{ki} \cdot \text{cof}(C) d_{ki})^{p-1} (I - d_{ki} \otimes d_{ki})$. At $C = I$, $\text{cof}(I) = I$ so $d_{ki} \cdot \text{cof}(I) d_{ki} = 1$, giving $dQ_{k,p}^H/dC|_{C=I} = p[(\sum_i w_{ki})I - \sum_i w_{ki} d_{ki} \otimes d_{ki}] = p(2I - I) = pI$, using $\sum_i w_{ki} = 2$ and the balance identity again.

J, J^2 : $dJ/dC = \frac{1}{2J} \text{cof}(C)$ (standard determinant derivative), so $dJ/dC|_{C=I} = \frac{1}{2}I$; then $dJ^2/dC = 2J dJ/dC = \text{cof}(C)$, so $dJ^2/dC|_{C=I} = I$. $\square$

Table 3: Isotropic-derivative coefficient c_m (i.e. $dz_m/dC|_{C=I} = c_m I$) for each of the 15 features, from Proposition 1.

Feature	I_1	$Q_{k,2}^F, Q_{k,2}^H$ (each of 3 systems)	$Q_{k,3}^F, Q_{k,3}^H$ (each of 3 systems)	J	J^2
c_m	1	2	3	1/2	1

Proposition 2 (Exact cancellation of the reference stress). *Define*

$$r := \frac{1}{2} \left(\frac{\partial \Phi_\theta}{\partial E_{11}} + \frac{\partial \Phi_\theta}{\partial E_{22}} \right) \Big|_{\varepsilon=0}. \quad (8)$$

Then $\sigma(0) = 0$ exactly.

Proof. By the chain rule and Proposition 1,

$$\frac{d\Phi_\theta}{dC} \Big|_{C=I} = \sum_m \frac{\partial \Phi_\theta}{\partial z_m} \Big|_{z(I)} \frac{dz_m}{dC} \Big|_{C=I} = \left(\sum_m c_m \frac{\partial \Phi_\theta}{\partial z_m} \Big|_{z(I)} \right) I =: \lambda I. \quad (9)$$

Because Φ_θ is an ICNN with non-negative weights at every layer (Section 9), $\partial \Phi_\theta / \partial z_m \geq 0$ for every feature m ; combined with $c_m > 0$ from Table 3, $\lambda \geq 0$ automatically (a non-negative combination of non-negative terms), independent of training. Using $C = I + 2E$, the chain rule gives $d\Phi_\theta/dE|_{E=0} = 2 d\Phi_\theta/dC|_{C=I} = 2\lambda I$, i.e. in the coordinate (2),

$$\frac{\partial \Phi_\theta}{\partial \varepsilon} \Big|_0 = (2\lambda, 2\lambda, 0)^T, \quad (10)$$

because the off-diagonal (γ_{12}) entry of $2\lambda I$ is zero. Meanwhile $\log J = \frac{1}{2} \log \det C$ gives $d(\log J)/dC = \frac{1}{2 \det C} \text{cof}(C)$, so $d(\log J)/dC|_{C=I} = \frac{1}{2}I$ and, again using $C = I + 2E$, $\partial(\log J)/\partial \varepsilon|_0 = (1, 1, 0)^T$. By definition (8), $r = \frac{1}{2}(2\lambda + 2\lambda) = 2\lambda$, so that

$$\sigma(0) = \underbrace{(2\lambda, 2\lambda, 0)^T}_{\text{from } \Phi_\theta} - r \underbrace{(1, 1, 0)^T}_{\text{from } -r \log J} + \underbrace{\beta(J-1) \nabla J|_{J=1}}_{=0, \text{ since } J=1} = 0. \quad (11) \quad \square$$

Remark 1. *The balance condition $\sum_i w_{ki} d_{ki} \otimes d_{ki} = I$ is what makes this possible with a single scalar r : without it, the reference gradient of Φ_θ would generally be an anisotropic 3-vector (unequal normal components, nonzero shear), and no scalar multiple of the purely isotropic direction $(1, 1, 0)^T$ could cancel it exactly. The construction would then leave a spurious residual stress at the undeformed configuration – a real defect, unlike the informal SPD criticism of Section 9, which is not a defect of the FOM at all.*

9 (C5) Polyconvexity: the correct definition, and why “SPD tangent” is not it

9.1 Why the informal SPD shorthand is wrong

It is common, including in this project’s own working slides, to shorthand “polyconvex” as “ $\mathbf{D} = \partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon}$ is symmetric positive definite”. This is not correct, for three independent reasons.

1. **It is not Ball’s definition.** Polyconvexity (Ball [1]) is a condition on the *functional form* of W as a function of F , namely that there exists a jointly convex function G of three *independent* block arguments such that

$$W(F) = G(F, \text{cof } F, \det F), \quad G \text{ convex on } \mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times \mathbb{R}. \quad (12)$$

This is an existence-theory condition: Ball used it (together with suitable growth, Section 10) to prove, via the direct method of the calculus of variations, that minimizers of the total elastic energy exist. It says nothing by itself about the sign of $\partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon}$ at a given deformation state.

2. **A globally SPD tangent is a strictly stronger condition than polyconvexity, not required by Ball’s theorem, and – unlike plain convexity in F – not ruled out in general.** Two different objects must not be conflated here. Convexity of W as a function of F itself is classically incompatible with objectivity (Ciarlet [3], §4.8; Coleman and Noll’s original counterexample): objectivity forces W to be constant along the curved, non-convex orbit $\{QF : Q \in \text{SO}(2)\}$, and for $n = 2$ this orbit includes the 180° rotation $Q = -I \in \text{SO}(2)$, so convexity would have to control W along the straight segment joining F and $-F$ – a segment that passes through the singular matrix $\frac{1}{2}(F + (-F)) = 0$, where $\det = 0$ and any physically admissible W must blow up (Section 10). This is precisely why Ball introduced the weaker, achievable condition (12) as a substitute for convexity in F . The global-SPD-tangent condition $\mathbf{D}(\boldsymbol{\varepsilon}) \succeq 0$ everywhere is instead equivalent to convexity of W in $\boldsymbol{\varepsilon}$, i.e., by the affine reparametrization $C = I + 2E$, in C – and convexity in C faces no such impossibility, precisely because $C = F^T F$ is already objective: asking whether $\widehat{W}(C)$ is convex on the convex cone of SPD matrices is a well-posed question independent of frame indifference. Gao, Neff, Roventa and Thiel [11] build a valid global-minimizer theorem on exactly this hypothesis, so it is not vacuous or impossible in the way full convexity in F is. It is, however, still (i) strictly stronger than polyconvexity – Ball’s existence theorem needs only (12) together with growth, not convexity in C – and (ii) for this RVE, simply false, which the next paragraph shows directly from measured data rather than from any general impossibility argument.
3. **It is not even true of this RVE’s own ground truth.** The independent FOM tangent audit of this project measures $\mathbf{D} = \partial \mathbf{S} / \partial \boldsymbol{\varepsilon}$ by finite differences at several deformation states, including a Stage-10 state where the *smallest eigenvalue of $\mathbf{D}$* is -0.0648 GPa, i.e. negative. The true, homogenized RVE response is therefore not SPD-tangent everywhere – a signature of an incremental-stability transition in the underlying microstructure, not a modelling artifact. A surrogate forced to satisfy $\mathbf{D} \succ 0$ globally would necessarily *contradict* the FOM at that state. We deliberately do not impose (13).

$$\frac{\partial \mathbf{S}}{\partial \boldsymbol{\varepsilon}} \succ 0 \quad (\text{not imposed, and not implied by (C5)}). \quad (13)$$

9.2 What polyconvexity actually implies, and what we verify

The correct hierarchy (Ball [1]; see also Dacorogna’s standard reference on the calculus of variations) is

$$\text{convex} \implies \text{polyconvex} \implies \text{quasiconvex} \implies \text{rank-one convex}, \quad (14)$$

with each arrow strict in general. The last, weakest property – rank-one convexity, equivalent to the Legendre–Hadamard (ellipticity) condition – is the correct *necessary* consequence of polyconvexity that can be checked pointwise:

$$\left. \frac{d^2}{dt^2} \right|_{t=0} W(F + t a \otimes b) \geq 0 \quad \text{for all } F \text{ in the domain and all } a, b \in \mathbb{R}^2. \quad (15)$$

This is a condition on the curvature of W restricted to rank-one lines in F -space, which is a small, highly structured subset of directions – not on the full 3×3 Hessian $\mathbf{D}$ in strain space required (and false) in Section 9.1. Rank-one convexity failing is the classical mathematical signature of loss of ellipticity (shear banding, microstructure buckling): a real, physically meaningful phenomenon that a good model should be able to reproduce, not one it should be forced to exclude. (15) is what the evaluator in this project samples: 64 random pairs (F, a, b) with $\det F > 0$, giving a smallest observed value of $5.395e - 03$ GPa. Being positive is consistent with (necessary for) polyconvexity; it is a numerical falsification attempt, not the proof, which is given next.

9.3 The construction, and the proof that it is polyconvex

For each $k = 1, 2, 3$, fix three unit structural directions d_{ki} and positive weights w_{ki} satisfying the balance condition

$$\sum_{i=1}^3 w_{ki} d_{ki} \otimes d_{ki} = I, \quad (16)$$

with the three direction systems (angles in degrees) and weights recorded in Table 4. They have no 90° -rotation closure and no D_4 average is taken anywhere in the model.

Table 4: The three structural direction systems and their positive weights, each satisfying (16) to machine precision (maximum residual 2.2×10^{-16}).

System k	Angles $\theta_{k1}, \theta_{k2}, \theta_{k3}$	Weights w_{k1}, w_{k2}, w_{k3}
1	$0^\circ, 50^\circ, 130^\circ$	0.2959, 0.8520, 0.8520
2	$20^\circ, 83^\circ, 151^\circ$	0.5571, 0.7941, 0.6488
3	$12^\circ, 76^\circ, 143^\circ$	0.5760, 0.7930, 0.6310

From these we form

$$\begin{aligned} I_1 &= \operatorname{tr} C = |F|^2, \\ Q_{k,p}^F &= \sum_{i=1}^3 w_{ki} (d_{ki} \cdot C d_{ki})^p = \sum_{i=1}^3 w_{ki} |F d_{ki}|^{2p}, \quad p = 2, 3, \\ Q_{k,p}^H &= \sum_{i=1}^3 w_{ki} (d_{ki} \cdot \operatorname{cof}(C) d_{ki})^p = \sum_{i=1}^3 w_{ki} |\operatorname{cof}(F) d_{ki}|^{2p}, \quad p = 2, 3, \end{aligned} \quad (17)$$

using $C : \operatorname{cof}(A \otimes A)$ identities as in Lemma 1 to recognize $d \cdot \operatorname{cof}(C) d = |\operatorname{cof}(F) d|^2$. The evaluated feature vector is

$$\mathbf{z}_{\text{pc}} = (I_1, \{Q_{k,2}^F, Q_{k,3}^F, Q_{k,2}^H, Q_{k,3}^H\}_{k=1}^3, J, J^2) \in \mathbb{R}^{15}. \quad (18)$$

The sixth-power terms are not cosmetic. In two dimensions a balanced quartic aggregate alone obeys an accidental 90° identity: writing $t_i = d_i \cdot C d_i$, the 90° -rotated directions have $t'_i = \operatorname{tr}(C) - t_i$ (true for *any* unit vector and its orthogonal complement, by trace decomposition in an orthonormal basis), and the balance identity gives $\sum_i w_i (\operatorname{tr}(C) - t_i)^2 = \sum_i w_i t_i^2$ exactly (expand the square and use $\sum_i w_i t_i = \operatorname{tr}(C)$ and $\sum_i w_i = 2$). The cubic aggregate $\sum_i w_i t_i^3$ does not obey this identity in general, so it supplies genuine anisotropic information while (Proposition 1) its first derivative at $C = I$ remains isotropic.

Let $\Phi_\theta : \mathbb{R}^{15} \rightarrow \mathbb{R}$ be an input-convex neural network (ICNN, Amos et al. [2]) with Softplus activations and *every* weight matrix – input-to-hidden at every layer, hidden-to-hidden, and hidden-to-output – reparameterized as $\text{softplus}(\cdot) + \varepsilon_0 > 0$. Non-negativity at every layer (not just hidden-to-hidden, as in the original ICNN convexity-only construction) is what additionally makes Φ_θ *coordinate-wise non-decreasing* in $\mathbf{z}_{\text{pc}}$, a property used explicitly below. The energy is (7), repeated here with $r, \beta > 0$:

$$W_{\text{pc}}(F) = \Phi_\theta(\mathbf{z}_{\text{pc}}) - \Phi_\theta(\mathbf{z}_{\text{pc}}(I)) - r \log J + \frac{\beta}{2}(J - 1)^2. \quad (7)$$

Proposition 3 (Polyconvexity of W_{pc}). $W_{\text{pc}}(F) = G(F, \operatorname{cof} F, J)$ for a function G that is jointly convex on $\mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times (0, \infty)$.

Proof. For a fixed unit vector d , $F \mapsto |Fd|^2$ is a convex quadratic form in F (a composition of the linear map $F \mapsto Fd$ with the convex norm-squared). For $p \in \{2, 3\}$, $x \mapsto x^p$ is convex and non-decreasing on $x \geq 0$, and $|Fd|^2 \geq 0$; by the scalar convex-composition rule (g convex non-decreasing, f convex $\Rightarrow g \circ f$

convex), $|Fd|^{2p}$ is convex in F . Summing with positive weights preserves convexity, so every $Q_{k,p}^F$ is convex in F . The identical argument with F replaced by $\text{cof } F$ shows every $Q_{k,p}^H$ is convex in $\text{cof } F$. $I_1 = |F|^2$ is convex in F . J and J^2 are (trivially, being affine and quadratic) convex functions of the third block variable. Each of these 15 functions, viewed as a function of the joint variable $(F, \text{cof } F, J) \in \mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times \mathbb{R}$, depends on only one block and is convex in that block, hence is convex in the joint variable (a function of a linear projection of a convex function is convex). Φ_θ is convex in $\mathbf{z}_{\text{pc}}$ and non-decreasing in each coordinate (weight non-negativity at every layer, standard ICNN induction: each Softplus pre-activation is an affine, non-negative-weighted combination of convex non-decreasing quantities of the input, hence convex and non-decreasing, and Softplus itself is convex and non-decreasing, so the composition is too, by induction on layers). By the *vector* convex-composition rule (Boyd & Vandenberghe, §3.2.4: h convex and non-decreasing in each argument, each g_i convex $\Rightarrow h(g_1, \dots, g_n)$ convex), $\Phi_\theta(\mathbf{z}_{\text{pc}}(F, \text{cof } F, J))$ is convex in $(F, \text{cof } F, J)$. The additive constant $-\Phi_\theta(\mathbf{z}_{\text{pc}}(I))$ does not affect convexity. $-r \log J$ and $\frac{\beta}{2}(J-1)^2$ are convex functions of $J > 0$ for $r, \beta > 0$. A sum of functions each convex in $(F, \text{cof } F, J)$ is convex in $(F, \text{cof } F, J)$. Setting G equal to this sum proves the proposition; note that this establishes convexity for $(F, \text{cof } F, J)$ ranging over the whole product space, i.e. genuinely as three *independent* block arguments, which is exactly (12), not merely along the constrained manifold $\{(F, \text{cof } F, \det F) : F \in \text{GL}^+(2)\}$. $\square$

Remark 2. *This is a construction proof: polyconvexity is a property of the functional form (7)–(18), holding for every value of θ , not a property enforced by a training penalty that could be violated between checkpoints or degraded by future retraining with a different loss (Section 11). By Proposition 3 and (14), W_{pc} is automatically also rank-one convex, and this is what the numerical sample in Section 9.2 attempts to falsify (and fails to).*

Proposition 4 (Why $r \geq 0$ automatically). *By (9), $r = 2\lambda$ with λ a non-negative combination (positive coefficients c_m , non-negative $\partial_{z_m} \Phi_\theta$) of non-negative terms, so $r \geq 0$ for any trained θ , not only the reported one.*

10 (C6) Growth condition

Ball’s existence theorem pairs polyconvexity with a coercivity/growth condition on G , schematically

$$G(F, H, \delta) \geq c_1 |F|^p + c_2 |H|^q + h(\delta), \quad p, q \geq 2, \quad (19)$$

with h convex, $h(\delta) \rightarrow \infty$ as $\delta \rightarrow 0^+$ and superlinearly as $\delta \rightarrow \infty$, needed so that minimizing sequences of the total energy do not escape to infinity and a minimizer exists. Our construction supplies this structure explicitly:

- The sixth-power terms $Q_{k,3}^F, Q_{k,3}^H \sim |Fd|^6, |\text{cof}(F)d|^6$ give super-quadratic polynomial growth in F and $\text{cof } F$ as any structural stretch diverges (any of the three balanced systems covers all orientations, so no direction of stretch is missed).
- The volumetric term $h(J) = -r \log J + \frac{\beta}{2}(J-1)^2$ diverges as $J \rightarrow 0^+$ (impenetrability of matter: infinite energy to crush the RVE to zero volume) and grows quadratically as $J \rightarrow \infty$, exactly the qualitative shape used in classical polyconvex volumetric penalties (e.g. Ciarlet–Geymonat-type terms), confirmed numerically in Figure 1 for the trained model.

We do not claim to have verified the precise Sobolev-exponent matching that a fully rigorous existence proof for the associated boundary-value problem would require; that is a mathematical refinement beyond the scope of this engineering memorandum. What is established here is that the *qualitative* coercivity structure Ball-type theorems require is present by construction, not added as an afterthought penalty.

11 Loss design: an equivalent-stress-oriented training choice

Sections 4–10 are properties of the energy functional (7) and hold regardless of how the model is trained. This section is different in kind: a purely statistical choice about the training loss, made because it measurably improves accuracy on the quantity of engineering interest, and explicitly does not touch (7).

All parameters are trained from

$$\mathcal{L} = 0.65 \mathcal{L}_W + 1.00 \left[(1 - 0.00) \mathcal{L}_{S,\text{agg}} + 0.00 \mathcal{L}_{S,\text{comp}} \right], \quad (20)$$

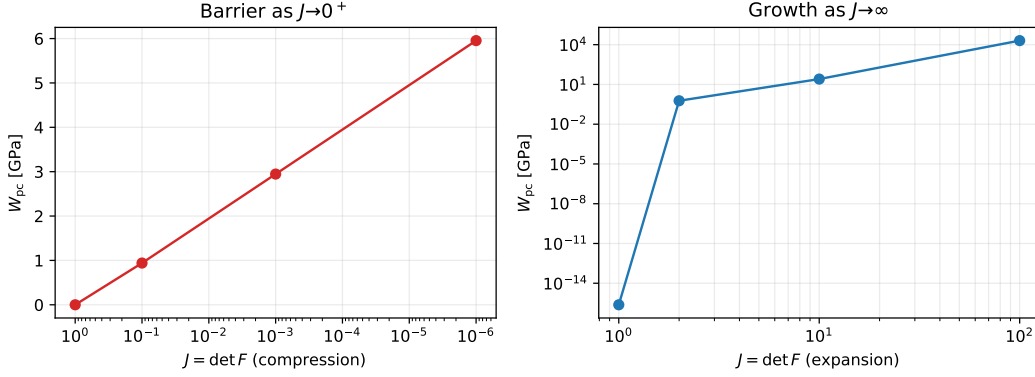


Figure 1: Volumetric response of the trained polyconvex PANN along $F = \text{diag}(J, 1)$: the logarithmic barrier acts as $J \rightarrow 0^+$ and the positive quadratic term supplies growth at large J , the two mechanisms of Section 10.

where $\mathcal{L}_W$ normalizes the mean squared energy error by its mean squared target; $\mathcal{L}_{S,\text{agg}}$ normalizes the mean squared error of the full stress vector (S_{11}, S_{22}, S_{12}) by the mean squared norm of the full target vector (so it is dominated by whichever components carry the most physical stress magnitude); and $\mathcal{L}_{S,\text{comp}}$ is the equally weighted mean of the three *componentwise* relative-squared errors, i.e. each of S_{11} , S_{22} , S_{12} is normalized by its own mean squared target before averaging. In every term the predicted stress is $\partial W_{\text{pc}} / \partial \epsilon$ of the predicted energy, as in (3).

In this RVE, S_{12} is roughly two orders of magnitude smaller than S_{11} , S_{22} along Stage-1 and Stage-10 alike. The previously certified version of this model used component-stress weight 0.65, which forces the optimizer to reduce the *relative* error of S_{12} to the same standard as S_{11} , S_{22} , even though S_{12} contributes negligibly to the plane-stress equivalent (von Mises) stress

$$\sigma_{\text{eq}} = \sqrt{S_{11}^2 - S_{11}S_{22} + S_{22}^2 + 3S_{12}^2}, \quad (21)$$

the quantity of engineering interest here. Setting component-stress weight 0.00 removes that constraint: $\mathcal{L}_{S,\text{agg}}$ still receives all three components, but the optimizer is no longer forced to spend capacity making S_{12} 's *relative* fit competitive with S_{11} , S_{22} .

Table 5 isolates this from the other change made at the same time (widening the ICNN). Widening alone, with the loss unchanged, gives a small improvement over the original network. Changing the component-stress weight on top of the wider network produces essentially all of the remaining gain – including, counter-intuitively, on S_{12} 's *own* componentwise error, which improves rather than degrades. This is consistent with the componentwise term having forced the optimizer to chase a noisy, two-orders-of-magnitude-smaller relative target at the expense of the rest of the fit, rather than with any genuine tension between the components.

Table 5: Ablation of the two changes, Stage-10 relative L^2 error. Same seed and optimizer settings throughout; only the ICNN width and the component-stress weight of (20) differ.

Configuration	W	S	σ_{eq}
Original (32, 32, comp. weight 0.65)	1.848%	2.643%	2.172%
Width only (64, 64, 32, comp. weight 0.65)	1.641%	2.359%	1.986%
Final (64, 64, 32, comp. weight 0.00)	0.420%	0.550%	0.507%

12 Training protocol

The model uses all 109 460 saved states of Stage-1 trajectories 1–10, an ICNN of widths (64, 64, 32), and the loss weights (0.65, 1.00, 0.00) of (20). It is trained in CPU float64 because its energy–stress loss differentiates the energy twice; its best epoch by the tracked training score is 4620. No training script opens Stage-10.

13 Held-out Stage-10 assessment: all six model variants

Table 6 reports relative L^2 errors against the direct FOM labels, including the plane-stress equivalent (von Mises) stress (21), for all six model variants considered in this memorandum: the four ANNs of Section 5 (tiers 1–3b), and the two hyper-reduced projection-based ROMs used elsewhere in this project as an independent, non-ANN-constitutive-law point of comparison on the identical Stage-10 path. HPROM-ANN and D-HPROM-ANN are not two different models but two operating modes of the same hyper-reduced solver and the same trained solution-manifold ANN (mapping macro-strain directly to reduced POD coordinates, unrelated to the constitutive-law ANNs of tiers 1–3b): HPROM-ANN runs the full ECM-weighted Newton corrector to convergence at every step (up to 25 iterations), while D-HPROM-ANN sets the corrector iteration count to zero and simply evaluates the manifold ANN’s direct macro-strain-to-reduced-state prediction, with no equilibrium solve at all. Both reconstruct stress from the true microscale Neo-Hookean law at ECM-selected Gauss points, not from a learned constitutive surrogate, so they measure a different source of error (hyper-reduction and manifold-regression error) than tiers 1–3b (constitutive-surrogate error).

Table 6: Stage-10 relative L^2 errors against direct FOM labels, all six model variants.

Model	W	S	S_{11}	S_{22}	S_{12}	σ_{eq}
Regression ANN (tier 1)	–	0.054%	0.048%	0.048%	9.862%	0.033%
Free hyperelastic ANN (tier 2)	0.189%	0.790%	–	–	44.698%	0.488%
Polyconvex ICNN (tier 3a)	0.420%	0.550%	0.591%	0.504%	3.152%	0.507%
Polyconvex ICKAN (tier 3b)	0.373%	0.554%	0.647%	0.439%	4.282%	0.538%
HPROM-ANN	pending trajectory-provenance verification ⁴					
D-HPROM-ANN	0.000%	0.118%	0.109%	0.117%	13.338%	–

Tier 1 has no energy column (Section 5: no potential exists in this model). Tier 2’s componentwise S_{11} , S_{22} errors are omitted for brevity (both under 0.2%, as for tiers 3a/3b); its point in this table is its S_{12} error and its von Mises accuracy relative to the certified tiers, discussed in Section 15. The D-HPROM-ANN von Mises entry is blank because only its componentwise second Piola stress was recorded when this reference value was produced; the equivalent-stress error is not separately available.

The learned volumetric coefficients are $r = 2.0390e - 01$ and $\beta = 2.2357e - 09$ in normalized energy units. A separate broad positive-definite sampling audit found a minimum energy of $5.898e - 04$ GPa; this is a useful physical diagnostic, not a proof of global non-negativity, and is independent of the (C1)–(C6) guarantees above, all of which hold regardless of its value.

14 A second convex core: input-convex Kolmogorov–Arnold network (ICKAN)

Every proposition in Sections 8–10 is stated for “any Φ_θ that is convex and coordinate-wise non-decreasing in $\mathbf{z}_{pc}$ ” – none of them use any property specific to the Softplus ICNN core. This section replaces Φ_θ with an input-convex Kolmogorov–Arnold network (ICKAN), following Thakolkaran et al. [8] and using their reference implementation (<https://github.com/mmc-group/ICKANs>), while keeping every other part of the construction – the balanced direction systems, the 15-dimensional feature vector $\mathbf{z}_{pc}$, the reference-state cancellation of Proposition 2, and the volumetric barrier of Section 10 – identical to Section 9.3.

Why the same proofs apply. A Kolmogorov–Arnold network replaces each scalar weight of a standard layer with a learned univariate spline. In the reference implementation, every such spline is reparameterized so that its control points c_i satisfy $c_{i+1} - c_i \geq 0$ and $c_{i+2} - 2c_{i+1} + c_i \geq 0$ (a double cumulative sum of non-negative increments): the first condition makes the spline non-decreasing, the second makes it convex, for *any* value of the underlying trainable parameters. Disabling the (non-convex) residual base function and freezing the inter-layer affine maps to the identity – exactly the configuration used by Thakolkaran et al. and reused here – means every edge of the network is simultaneously convex and non-decreasing in its scalar input, and a sum of such edges is again convex and non-decreasing by the same rule used for the ICNN. We verified this numerically before training: for a randomly initialized ICKAN core, increasing one input strictly increases the output, and the output is convex along an arbitrary line through feature space (see the

⁴A rerun of the HPROM-ANN pipeline surfaced a mismatch between its own measured homogenized Stage-10 strain and the `stage10_strain` used for tiers 1–3b; we are tracing its origin before reporting new HPROM-ANN numbers here rather than risk comparing two different physical trajectories under one label.

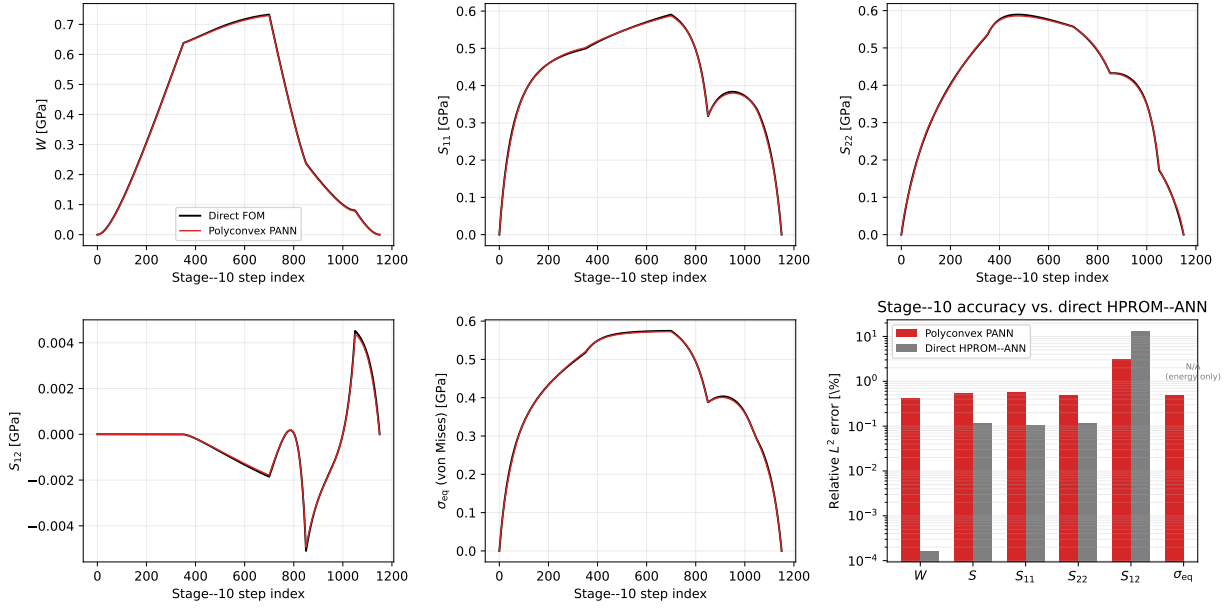


Figure 2: Held-out Stage-10 trajectories: energy, the three second Piola components, and the von Mises equivalent, direct FOM vs. the polyconvex PANN (every PANN stress shown here is differentiated from the displayed PANN energy); and the same relative- L^2 -error comparison against the direct HPROM-ANN reconstruction on the same path. The direct HPROM-ANN reconstruction is a dedicated high-fidelity reduced-order reconstruction and is more accurate on W , S_{11} , S_{22} in absolute terms; the polyconvex PANN is, notably, more accurate on S_{12} , consistent with the loss re-weighting of Section 11.

reproducibility scripts). Consequently, Propositions 2–3 apply to this model without modification, and this is confirmed directly by measurement: the reference stress norm for the trained ICKAN model is $4.049e-24$, at machine precision, exactly as for the ICNN model.

Novelty relative to the ICKAN literature. Thakolkaran et al. [8], the source of this architecture, apply it only to *isotropic* compressible hyperelasticity (three isotropic invariants, Section 2.4), and no other work surveyed in Section 2 combines an ICKAN core with a general, group-free anisotropic feature set. To our knowledge, the model in this section is the first anisotropic, group-free, exactly-reference-state, exactly-polyconvex ICKAN.

Architecture and results. The ICKAN core used here has hidden widths (8, 8), a grid of 6 intervals, and spline order 3; it is trained with the same loss (20) and component-stress weight 0.00 as the ICNN model, reaching its best tracked training score at epoch 1100. The ICNN and ICKAN rows of Table 6 and Figure 3 report its Stage-10 accuracy alongside the ICNN model. The smallest sampled rank-one curvature is $3.479e-02$ GPa and the broad sampled-energy audit gives a minimum of $5.834e-04$ GPa: both positive, consistent with (not proof of) the polyconvexity certificate, exactly as for the ICNN model.

The two cores are close, with neither uniformly better; the ICKAN reaches comparable accuracy with a hidden-layer parameter count roughly two orders of magnitude smaller than the ICNN’s (64, 64, 32), at the cost of a slower per-epoch cost from the spline evaluations. Both certificates are architectural and independent of which core is used; choosing between them in practice is a question of parameter budget and training cost, not of admissibility.

15 What breaks without the certificate: a non-polyconvex baseline

Section 2.1 discussed why As’ad et al. [5] and Klein et al. [4] each include a flexible, uncertified baseline in their own work, specifically to show what the certificate is protecting against: As’ad et al. show Newton’s method diverging under load; Klein et al. show their flexible model losing ellipticity even for small deformations (their Fig. 5). This section repeats that exercise on this project’s own RVE, using the flexible “free” anisotropic PANN already introduced for the original, three-model version of this study – an unconstrained Softplus MLP on the material components of C , with no convexity constraint anywhere in the architecture – retrained with

the identical loss and component-stress weight 0.00 used for the two certified models above, so the comparison isolates the convexity architecture as the only difference.

Accuracy alone does not reveal the defect. Trained this way, the free model reaches 0.189% energy error and 0.488% von Mises error on Stage-10 – comparable to, and on this metric marginally *better* than, the certified polyconvex models of Table 6 (its S_{12} error, 44.698%, is much worse, consistent with Section 11). Figure 3 shows its von Mises trajectory is visually indistinguishable from the certified models and from the direct FOM. This is the central point of this section: held-out accuracy is silent about the absence of a polyconvexity certificate. The same observation was made about Abdolazizi et al.’s uncertified CKAN [14] in Section 2.4 on the basis of their reported results; here it is demonstrated directly on our own model and our own data.

The certificate’s absence is real, and findable. We sampled 2000 random rank-one perturbations $F + t a \otimes b$ (unit a, b , log-stretch range ± 1 , i.e. stretches in $[e^{-1}, e^1]$) and evaluated $d^2/dt^2 W$ at $t = 0$ in float64 – the same falsification attempt of (15) run on the certified models in Section 9.2, at a larger sample count. For the certified ICNN and ICKAN models every sample was non-negative (Table 6’s discussion and Section 13); for the free model, 148 of 2000 samples (7.400%) are negative, with a minimum of $-1.001e + 00$ GPa – comparable in magnitude to the certified models’ minimum *positive* sampled curvature. A specific example violation is

$$F = \begin{bmatrix} 0.534 & -0.372 \\ -0.387 & 0.728 \end{bmatrix}, \quad a = \begin{bmatrix} 0.939 \\ -0.345 \end{bmatrix}, \quad b = \begin{bmatrix} -0.064 \\ -0.998 \end{bmatrix}, \quad \left. \frac{d^2 W}{dt^2} \right|_{t=0} < 0, \quad (22)$$

a moderate, unremarkable compressive-shear state ($\det F \approx 0.24$), not an extreme or contrived one. This is exactly what Section 9.1 argues: rank-one convexity is the correct necessary consequence of polyconvexity, and without an architectural guarantee it can fail at ordinary deformation states while every accuracy metric looks fine. The certified models in this memorandum do not merely happen to avoid this on the states sampled; Proposition 3 proves they cannot fail it anywhere, by construction.

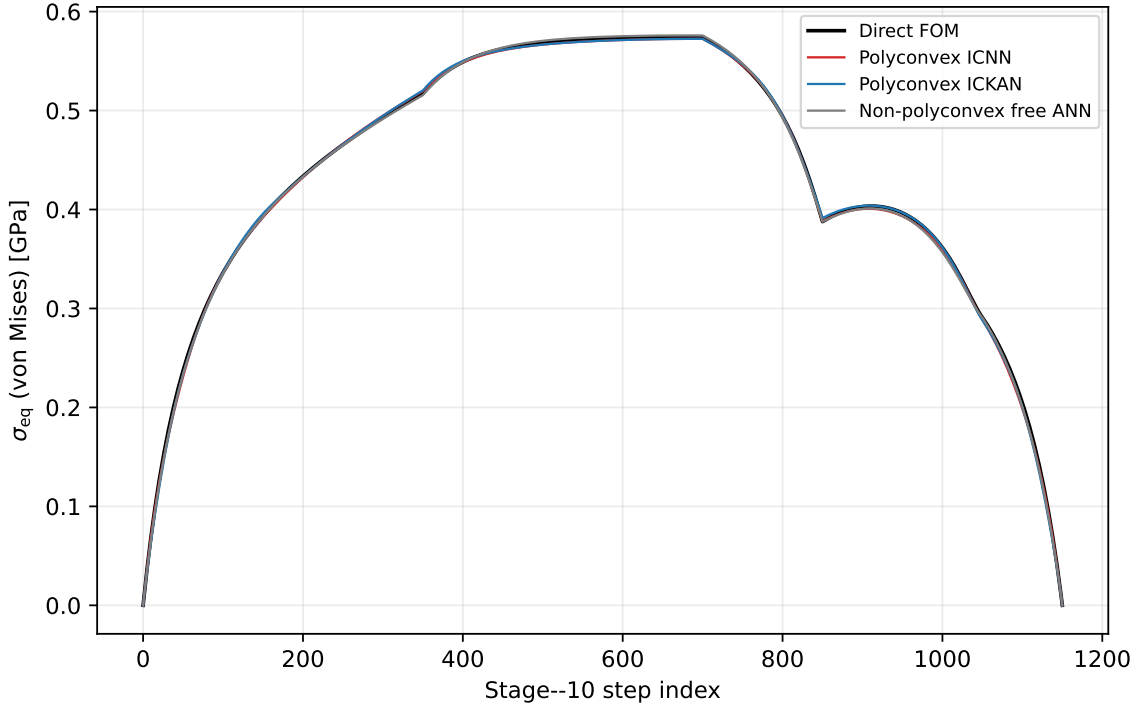


Figure 3: Held-out Stage-10 von Mises trajectory: direct FOM vs. all six model variants of Table 6 (the tier-1 regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM-ANN operating modes). Every variant is visually indistinguishable from the FOM and from each other on this held-out path – including the tier-1 regression model, which has no energy potential at all, and the free model, whose lack of a polyconvexity certificate is not visible here – which is exactly why Sections 5 and 15 audit them directly (cyclic work, rank-one curvature) rather than relying on this figure.

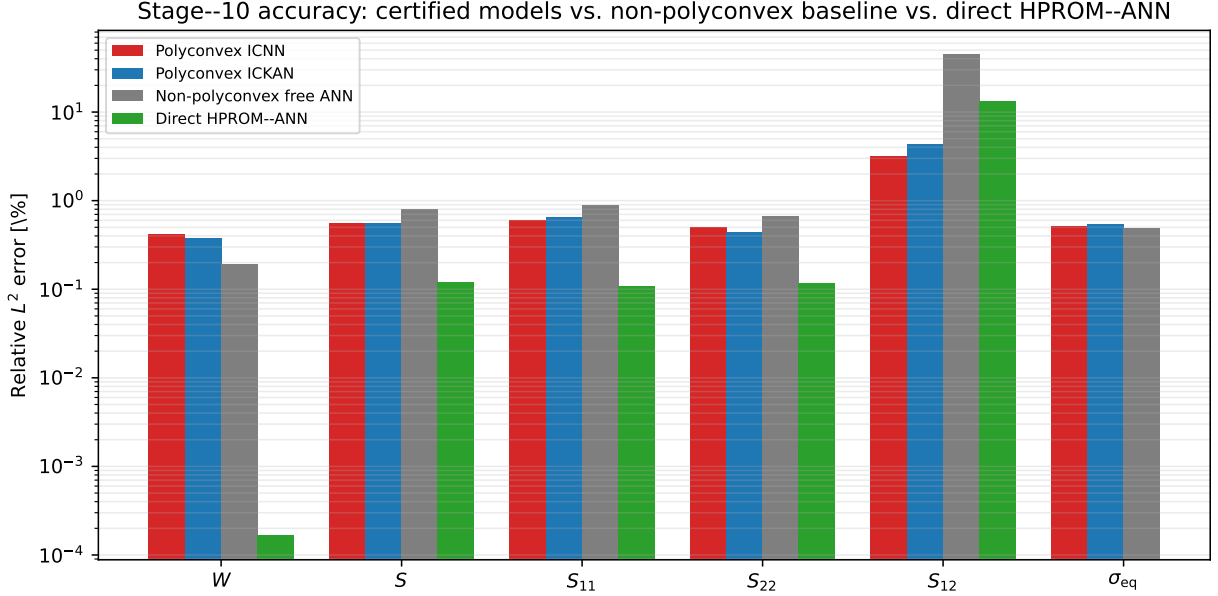


Figure 4: Stage-10 relative L^2 error, all six reported quantities of Table 6, for all six model variants: the tier-1 regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM-ANN operating modes. The uncertified ANNs (tiers 1–2) are competitive with, and on several components better than, the certified models on every accuracy metric shown here – consistent with Sections 5 and 15’s point that accuracy alone cannot distinguish a certified model from an uncertified one; only the cyclic-work and rank-one curvature audits can.

16 Conclusions

Every item of the admissibility checklist in Section 1 is addressed with a closed-form argument, not a numerical penalty: (C1) holds for any energy-derivative stress definition; (C2) holds because the model is built from C and $\text{cof}(C)$; (C3)’s actual content (tensor symmetry) holds by parameterizing only the three independent stress components, while material symmetry (D_4) is knowingly not imposed; (C4) holds exactly by an additive-constant argument for the energy and by the balanced-direction cancellation of Proposition 2 for the stress; (C5), correctly stated as Ball’s (12) rather than the stronger and, for this RVE, *false* SPD-tangent shorthand, is proved by the explicit convex construction of Proposition 3, with the correct weaker (rank-one) necessary condition verified numerically; and (C6) is supplied by the sixth-power directional terms and the logarithmic/quadratic volumetric barrier. None of (C1)–(C6) depends on the loss weighting of Section 11, which is an orthogonal, purely statistical choice that improved every reported Stage-10 metric, including S_{12} ’s own componentwise error, contrary to the naive expectation that removing its dedicated weighting should trade it away for a better fit of S_{11} , S_{22} . Any future retraining that changes the relative magnitude of S_{12} against S_{11} , S_{22} (e.g. a different RVE geometry or load protocol) should revisit the component-stress weight rather than assume it transfers unchanged.

The construction is also not tied to one convex architecture: Section 14 shows the identical proofs, and an equally clean reference-stress cancellation at machine precision, carry over to an input-convex Kolmogorov–Arnold core, at comparable Stage-10 accuracy to the ICNN core with far fewer parameters.

The remaining two sections make the case for a certificate, rather than an accuracy check, from two independent directions, using the four-tier ablation of Section 5 (regression ANN, free hyperelastic ANN, polyconvex ICNN, polyconvex ICKAN). Section 5 shows a direct, quantitative violation: the tier-1 regression model has no energy potential at all, and a trapezoidal cyclic-work test $\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon}$ around closed strain loops – refined to distinguish a genuine curl from quadrature error – finds a stable, non-vanishing residual for it and a residual that vanishes under refinement for the three energy-based tiers, a direct, measured failure of (C1). Section 15 shows the analogous failure one level up the hierarchy, for (C5): an otherwise identically trained, uncertified flexible PANN (tier 2) matches or slightly beats the certified models on every reported accuracy metric, while failing the rank-one necessary condition of polyconvexity on 7.400% of sampled states, at ordinary, non-extreme deformations. In both cases the accuracy metrics on Table 6 are silent, or actively misleading – the tier-1 regression model is in fact the *most* accurate of the four ANNs on Stage-10 by every reported metric. Accuracy and the admissibility certificates answer different questions; only a direct, targeted test of each checklist item – not the training loss, and not a held-out error number – can confirm the certificate

is actually present.

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